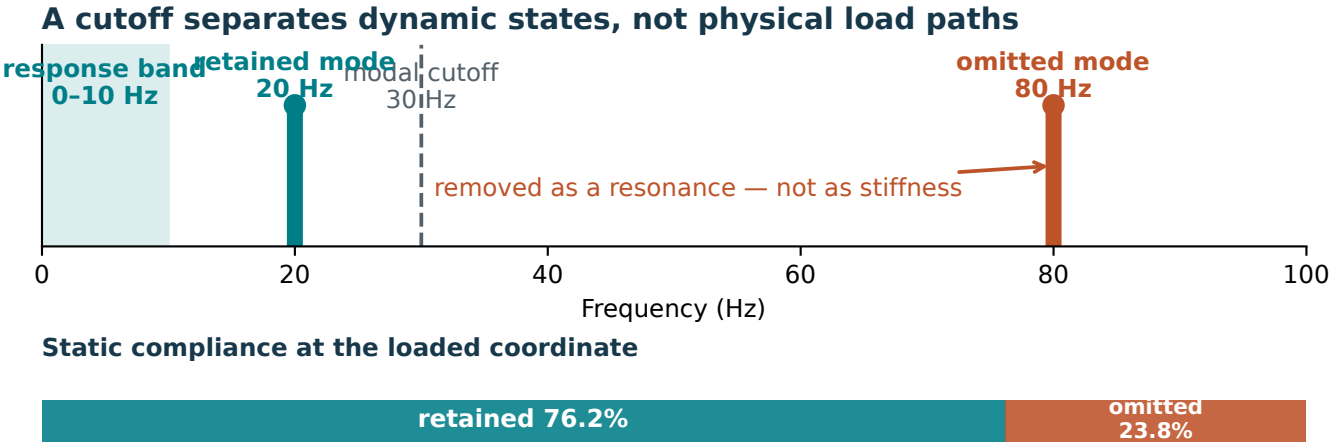
FINITE-ELEMENT PRACTICE + WORKED EXAMPLE / ABOUT 10 MINUTES

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The stiffness hiding above your modal cutoff



Illustrative two-mode FEM; percentages are computed below, not measured data.

Figure 1. Original conceptual illustration and calculation. A mode omitted from the dynamic basis can still carry static flexibility at the loaded coordinate. It is not test data and represents no particular vehicle.

A reduced FEM can look persuasive in the usual review plot. Its retained resonance peaks line up, the effective mass is high, and the analysis band sits comfortably below the last extracted mode. Then an interface force or local stress comes out unexpectedly low.

That is not a paradox. A modal cutoff removes dynamic coordinates, but it does not make the associated physical stiffness infinite. Modes above the cutoff still deform quasi-statically under lower-frequency load. NASA documented a sharp version of this problem in a 2024 IV&V study: a constrained reduction produced response below an inline analysis even though the selected cutoff followed the guidance in use. A sequence of Hurty–Craig–Bampton models converged only when the cutoff exceeded eight times the frequency of interest; an unconstrained mode-acceleration solution agreed with that converged result. That factor was case-specific, not a new rule. [2]

The objective today is to recognize modal truncation as both a dynamic-content and a static-flexibility problem; to calculate the missing term in a transparent FEM example; and to choose residual flexibility, residual vectors or mode acceleration with their limitations visible.

01 / WHAT ORDINARY MODAL SUPERPOSITION LEAVES BEHIND

The quiet modes still bend

For a linear FEM, let M and K be the mass and stiffness matrices and let the mass-normalized modes satisfy $\Phi^T M \Phi = I$. With proportional modal damping, a harmonic receptance assembled from the first N modes is

$$\mathbf{H}_N(\omega) = \sum_{r=1}^{N} \frac{\boldsymbol{\phi}_r \boldsymbol{\phi}_r^{\mathrm{T}}}{\omega_r^2 - \omega^2 + 2\mathrm{i}\zeta_r \omega_r \omega}$$

This basis reproduces the retained poles. At zero frequency, however, it supplies only $\Sigma \varphi_r \varphi_r{}^T/\omega_r{}^2$, while the exact static flexibility is K^{-1} after rigid-body and constraint handling. Their difference is the residual-flexibility matrix:

$$\mathbf{R} = \mathbf{K}^{-1} - \sum_{r=1}^{N} \frac{\boldsymbol{\phi}_r \boldsymbol{\phi}_r^{\mathrm{T}}}{\omega_r^2}$$

A static correction adds Rf to the retained modal displacement. In frequency-response form,

$$\mathbf{u}(\omega) \approx [\mathbf{H}_N(\omega) + \mathbf{R}]\,\mathbf{f}(\omega)$$

This is exact at zero frequency if R is built from the complete, consistently constrained model. It is an approximation away from zero because each omitted mode is being replaced by its static tail. The correction cannot reproduce an omitted resonance, a closely coupled nonlinear joint, or a load path that was absent from the parent FEM.

Mode acceleration expresses the same static recovery in a useful time-domain form:

$$\mathbf{u}_{\mathrm{MA}}(t) = \mathbf{K}^{-1}\mathbf{f}(t) - \sum_{r=1}^{N} \frac{\boldsymbol{\phi}_r \boldsymbol{\phi}_r^{\mathrm{T}} \mathbf{f}(t)}{\omega_r^2} + \sum_{r=1}^{N} \boldsymbol{\phi}_r q_r(t)$$

The first two terms are the residual static response; the last is the retained dynamic response. A NASA NASTRAN paper from 1992 notes the practical distinction: mode-displacement recovery can be efficient for accelerations when the modal basis envelopes the forcing, while mode acceleration is preferred when accurate displacements, element forces and stresses require the static contribution of truncated modes. It also warns that substructuring can complicate where that correction is applied. [4]

02 / A TWO-MODE FEM YOU CAN AUDIT BY HAND

Twenty-four percent of the compliance is invisible

Take an illustrative two-DOF, unit-mass model with orthonormal eigenvectors. At the collocated force/output coordinate, the modal residues are $a_1 = 1/6$ and $a_2 = 5/6$; the modal frequencies are 20 and 80 Hz, each with 1% damping. The full scalar receptance is

$$H(\omega) = \frac{a_1}{\omega_1^2 - \omega^2 + 2i\zeta_1\omega_1\omega} + \frac{a_2}{\omega_2^2 - \omega^2 + 2i\zeta_2\omega_2\omega}$$

Suppose the stated response band is 0–10 Hz and the reduced basis keeps only the 20 Hz mode. At zero frequency,

$$H_1(0) = 1.055 \times 10^{-5}\ \text{m/N} \qquad H_2(0) = 0.330 \times 10^{-5}\ \text{m/N}$$

The omitted 80 Hz mode carries 23.8% of the exact static receptance. A frequency-only argument misses it because the large residue offsets the larger denominator. Add $R = a_2/\omega_2^2$ and the static value is exact.

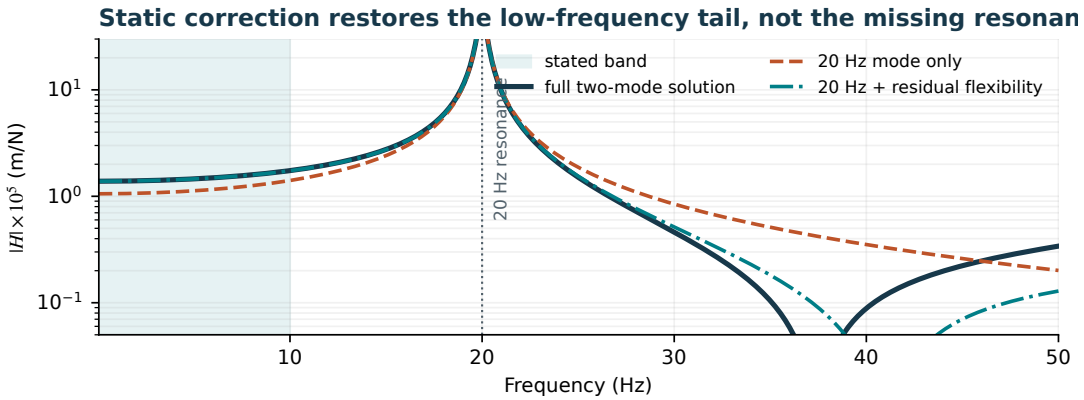


Figure 2. Original calculation from the disclosed two-mode model. The residual term restores the low-frequency tail but deliberately does not recreate the 80 Hz pole.

Frequency	20 Hz mode only	With residual
0 Hz	−23.8%	0.0%
5 Hz	-22.7%	-0.09%
10 Hz	-19.2%	-0.30%
30 Hz	+83.2%	+11.7%

Table 1. Magnitude error relative to the full solution. Static correction is excellent inside the stated band, then loses authority as frequency approaches the omitted dynamics and cancellations become important. These values are illustrative, not a criterion.

03 / BUILD THE CORRECTION FROM THE LOADS YOU ACTUALLY HAVE

Residual vectors are load-path probes

A full residual-flexibility matrix is usually too large to carry into a reduced model. Instead, solve a set of static load cases F, remove the flexibility already represented by retained modes, and use the remaining displacement patterns as additional basis vectors:

$$\mathbf{V}_{\text{res}} = \mathbf{K}^{-1}\mathbf{F} - \boldsymbol{\Phi}_N \boldsymbol{\Omega}_N^{-2} \boldsymbol{\Phi}_N^{\mathrm{T}} \mathbf{F}$$

Then orthogonalize, normalize and check independence before appending those vectors to the basis. One vector per loaded degree of freedom can be expensive; a smaller set of physically representative force patterns can be efficient but only spans those patterns. Solver implementations differ. For example, Altair’s OptiStruct documentation distinguishes unit-load, applied-load and enforced-displacement methods and describes their accuracy/cost trade. Treat those as implementation guidance, not a universal aerospace rule. [5]

A disciplined finite-element workflow is short enough to put in the analysis plan:

1 — Define the requested outputs. Interface forces, element stresses and relative displacements are often more sensitive to missing flexibility than a handful of accelerations. Include every operational load pattern, enforced motion and recovery location that matters.

2 — Create a full-model benchmark. Use direct frequency response, direct transient response, or a deliberately high-cutoff modal solution on a manageable set of cases. A reduced model cannot validate itself.

3 — Sweep the basis. Increase cutoff and compare both peaks and off-resonance response. Repeat with and without the residual treatment. Convergence should be judged on decision-driving outputs, not only frequency or effective-mass plots.

4 — Stress the boundaries. Vary the fixed-interface set, support stiffness, joint assumptions and load application. The 2024 NASA case is a reminder that a fixed-base constraint can suppress dynamic content before the free-free reduction ever sees it. [2]

5 — Audit the algebra. Verify units, mode normalization, rigid-body treatment, damping convention, recovery matrices and whether residual content survives substructuring and solver export. A residual vector tied to the wrong load basis is precision applied to the wrong problem.

04 / THE STANDARD SETS A FLOOR, NOT A CONVERGENCE PROOF

Read the “and,” not only the cutoff

NASA-STD-5002B, dated March 10, 2026, requires—for spacecraft and payload loads models within its adopted scope—a model upper bound and dynamic-reduction cutoff of at least 1.5 times the loads-analysis cutoff. Its next requirement says that residual vectors or mode acceleration shall capture the static-elastic contribution above the cutoff when calculating limit loads. In other words, frequency coverage and static recovery are separate obligations. [1]

That 1.5 factor is not a universal law of modal convergence. The standard applies to spacecraft and payload hardware when invoked by program or contract; it explicitly excludes launch vehicles, sounding-rocket payloads, aircraft, balloons and ground-support equipment from its scope. Program technical authority still defines the relevant analysis band and model fidelity. The 2024 IV&V example reached convergence only above eight times the band because of its particular reduction and constraints; exporting “8×” as a new heuristic would repeat the same mistake in the other direction. [2]

Residual content is not magic, either. Kammer and Blelloch’s NASA reanalysis study found a case where six interface residual vectors had little effect, while 192 vectors associated with a 32-node interface produced a far richer basis and changed a 287-mode prediction from very inaccurate to very accurate. In other formulations, residuals did not cure the error. The established finding is that residuals can profoundly reduce truncation error when their load/interface space matches the missing response. The hypothesis to test on your model is whether your chosen vectors span that space. [3]

Before release, ask for four plots: full versus reduced response at critical outputs; convergence versus cutoff; with-versus-without residual treatment; and sensitivity to constraint/interface selection. Add a static check: apply each correction load to the parent FEM and confirm the reduced basis reproduces the displacement and recovered force pattern. If the governing result changes materially when one vector is added, that is information about basis adequacy—not a nuisance to average away.

Finally, keep the physics boundary clear. Linear residual methods do not repair contact changes, preload-dependent joints, nonlinear damping, geometric stiffness omitted from K, or high-frequency inertia inside the analysis band. They recover missing linear flexibility under the modeled constraints and load subspace.

05 / LEAVE THE REVIEW WITH A TESTABLE CLAIM

Three takeaways

1. Modal cutoff is not a stiffness cutoff. Truncated modes can contribute materially to low-frequency displacement, stress and interface force even when their resonances are well above the band.

2. Frequency coverage and static recovery answer different questions. Retain enough modes to represent dynamics; then use residual flexibility, residual vectors or mode acceleration to recover the missing static-elastic tail where applicable.

3. Convergence belongs to outputs and load paths. Effective mass, mode count and a multiplier on frequency are screening evidence. Benchmark decision-driving responses against the parent model, vary constraints, and prove that the residual basis spans the applied loads.

A short reflection

A reduced model matches all retained eigenfrequencies and captures 95% effective mass in each translation. Can you conclude its bracket stress is converged?

Answer: No. Effective mass is a global participation measure, not a guarantee of local flexibility or stress recovery. Compare bracket stress against a higher-cutoff or direct solution, include its governing load pattern in the residual basis, and test sensitivity to interface constraints and recovery formulation.

Sources and figure notes

[1] NASA-STD-5002B, Load Analyses of Spacecraft and Payloads, March 10, 2026. Sections 1.2 and 4.4.2–4.4.3, especially LAR 26–29. Scope and contractual applicability matter.

[2] E. Simmons, Alternate Methods of Model Reduction to Avoid Dynamic Modal Truncation Error, NASA SLAMS Early Career Forum, July 23–25, 2024. NASA peer-committee review; public U.S. Government work.

[3] D. C. Kammer and P. A. Blelloch, An Examination of Launch Vehicle Loads Reanalysis Techniques, NASA/CR-2018-220091, September 2018. See the residual-vector comparisons in Sections 4–5.

[4] S. K. Thampi et al., “Transient Loads Analysis for Space Flight Applications,” in Twentieth NASTRAN Users’ Colloquium, NASA-CP-3145, April 1992, pp. 154–168.

[5] Altair, Modal Frequency Response Analysis, OptiStruct 2025 help. Official solver-specific descriptions of residual-vector options; not a program requirement.

Figures 1–2 and Table 1 are original illustrative calculations. No measured data, proprietary model or flight result is represented. This run could not verify or select GPT-6 Astra.